

# FedClinic-Fair: Transfer-Assisted Personalized Federated Learning for Client-Fair Medical Image Classification

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**Abstract**—Federated learning trains a shared model from data held by different clients, but client data are often non-identically distributed. In this setting, a model can have good average performance while performing poorly for some clients. This project studies whether lightweight personalization improves these weaker clients. We evaluate seven federated learning approaches on five simulated clients using PneumoniaMNIST, BreastMNIST, and BloodMNIST. The comparison includes established methods (FedAvg, FedProx, FedPer, and Ditto), a simple head fine-tuning baseline, a warm-start FedPer ablation that we call WarmFedPer, and a head-restricted proximal extension called ProxHead. All methods share a frozen ImageNet-pretrained ResNet-18 representation and train only a small adapter and classifier head. Hyperparameters are selected on a development seed using validation data, and final results use fresh paired seeds. Our main lower-tail metric is the average Macro-F1 of the two weakest clients. WarmFedPer gives the best average cross-dataset lower-tail rank. Compared with FedAvg, its bottom-two Macro-F1 improves by 0.1004 on PneumoniaMNIST and 0.0811 on BloodMNIST, while the BreastMNIST difference of 0.0205 is within seed noise. However, no single method is best on every dataset: Head-FT is strongest on several BreastMNIST and BloodMNIST metrics, while ProxHead gives the highest BloodMNIST bottom-two Macro-F1 and performs well under the strongest tested heterogeneity. The main conclusion is therefore simple: warm-started lightweight personalization is a useful overall strategy in this simulation, but stronger proximal constraints are helpful only in some conditions.

**Index Terms**—federated learning, personalized federated learning, non-IID data, lower-tail performance, transfer learning, MedMNIST, medical image classification

## I. INTRODUCTION

Federated learning (FL) allows several clients to train a model together without directly pooling their raw training data. FedAvg is the standard starting point: clients train locally and a server averages their model updates [14]. A major difficulty is statistical heterogeneity. Different clients may have very different class distributions, so a single global model can work well on average while performing poorly for some clients. FedProx addresses part of this problem by constraining local drift during training [15].

This issue is especially relevant in medical FL. The closest study to our setting is the recent MedMNIST comparison by Riyadi et al., which evaluates FedAvg, FedProx, FedSVRG,

and FedAtt under Dirichlet non-IID partitions on PneumoniaMNIST, DermaMNIST, and BloodMNIST [1]. That work focuses on global FL methods and reports client-averaged performance. It does not study personalized heads or directly report worst-client and lower-tail performance. Other medical FL studies have explored partial model sharing [2], transfer-assisted FL [6], resource-aware fairness [7], and personalized parameter-efficient medical FL [8]. FLamby also shows the importance of evaluating FL in realistic cross-silo healthcare settings rather than relying only on synthetic client splits [9].

Personalized FL gives each client some local parameters. FedPer keeps personalization layers local [16]. FedRep formalizes a shared representation with client-specific heads, although its sequential update rule differs from the FedPer-style training used in this project [3]. Ditto learns full personalized models while regularizing them toward a global model [4]. Work on q-FFL also shows that average accuracy can hide poor outcomes for weaker clients and motivates reporting the lower part of the client performance distribution [5].

Initialization and the way local adaptation is performed can also matter. Nguyen et al. show that pretrained initialization can change federated optimization behavior under heterogeneity [10]. Recent work on personalized fine-tuning studies how local adaptation can distort a globally learned representation [11]. Broader recent studies of foundation-model and LoRA-based FL also report that heterogeneity can interact strongly with parameter-efficient adaptation [12], [13]. These papers use different domains and setups from ours, but they support the need to test personalization rather than assume that it helps every client.

This issue also appeared during our own development experiments. FedPer improved average performance, but some weak clients became worse. That observation changed the focus of the project. Instead of asking only whether personalization raises average accuracy, we ask:

*Under statistically heterogeneous simulated medical clients, can lightweight personalization improve lower-tail client performance while maintaining strong average performance?*

We study this question on three MedMNIST tasks [17]. We use five simulated clients, class-conditional Dirichlet parti-

tions, a frozen ImageNet-pretrained ResNet-18 representation [18], and a small trainable adapter plus classifier head. The frozen backbone is the transfer-assisted component of our approach: rather than training a network from scratch, we reuse the ImageNet knowledge and train only a small adapter and classifier head. We use client-fair to mean that weaker clients are not left behind by a model that looks strong on average, so we report worst-client and bottom-two Macro-F1 rather than the mean alone; the clients are simulated partitions, so this is a performance-distribution criterion and not demographic fairness. We compare established FL methods with two project variants: WarmFedPer, which starts personalization from a trained FedAvg model, and ProxHead, which adds a proximal penalty only to the personalized classifier head.

The paper makes three modest contributions:

- 1) It evaluates personalization using mean, worst-client, and bottom-two-client performance instead of relying only on average performance.
- 2) It compares global and personalized FL methods with a compute-matched warm-start ablation, separating the effect of warm initialization from the effect of proximal regularization.
- 3) It confirms the main observations on three datasets with fresh seeds and adds heterogeneity, representation, calibration, and client-level checks.

The main finding is not that one new method wins everywhere. WarmFedPer is the most consistent lower-tail method across the three datasets, while ProxHead is more useful in some high-heterogeneity conditions. This leads to a careful conclusion: warm-started lightweight personalization is a strong overall choice in this simulation, but stronger regularization is not always necessary.

## II. RELATED WORK AND COMPARED METHODS

### A. Medical Federated Learning

Medical FL work has studied several ways to handle data heterogeneity and privacy. Riyadi et al. provide the closest direct benchmark to ours because they use MedMNIST with a Dirichlet split and compare multiple FL optimizers [1]. Their results support the importance of heterogeneity-aware optimization, but the study uses three clients, a shallow CNN trained from scratch, and client-averaged metrics. Our study instead asks what happens to weaker clients when lightweight personalization is added.

FLOP shares only part of a medical imaging model and keeps part of the network private at each client [2]. This is an important architectural precedent for our shared-adapter/private-head setup, although FLOP is mainly motivated by privacy and reports averaged client accuracy rather than lower-tail metrics. Haripriya et al. combine transfer learning with medical FL [6], while Suresh et al. combine personalization and LoRA for leukemia imaging [8]. Zhang et al. study fairness in healthcare FL from the perspective of unequal client computing resources [7]; our fairness-related question is different because we study unequal predictive

performance caused by data heterogeneity. FLamby provides a useful reminder that real cross-silo medical FL contains natural institutional shifts that are richer than our simulated Dirichlet label skew [9].

### B. Personalization and Lower-Tail Performance

FedPer separates shared and client-specific layers [16]. FedRep studies a closely related shared-representation/local-head design and shows that representation sharing can be effective under heterogeneous client data [3]. Ditto uses a proximal connection between personalized and global models and explicitly studies fairness as the spread of client performance [4]. q-FFL instead changes the global objective to give more weight to high-loss clients and reports lower-performing client groups directly [5]. These works motivate our decision to report worst-client and bottom-two Macro-F1.

We compare seven methods:

- **FedAvg**: all trainable adapter and head parameters are globally aggregated.
- **FedProx-style**: FedAvg with a proximal penalty on all trainable parameters.
- **FedPer**: the adapter is aggregated, while each client keeps its own classifier head.
- **Head-FT**: after FedAvg training, the adapter is frozen and each client fine-tunes only its classifier head.
- **Ditto-style**: each client keeps a full personalized adapter-plus-head model regularized toward the global model.
- **WarmFedPer**: our compute-matched ablation. It starts from the selected FedAvg checkpoint, initializes all local heads from the global head, continues shared-adapter/personal-head training for 10 personalization rounds, reuses the ProxHead-selected head learning rate, and uses no proximal penalty.
- **ProxHead**: our project extension. It uses the same warm start and personalization schedule as WarmFedPer but adds a proximal penalty only to the classifier head.

WarmFedPer is not presented as a new published FL algorithm. It is a controlled project ablation used to test whether the benefit of ProxHead comes from warm initialization or from the proximal head constraint.

## III. METHODOLOGY

### A. Datasets

We use the official MedMNIST v2 train, validation, and test splits for PneumoniaMNIST, BreastMNIST, and BloodMNIST [17]. The first two are binary tasks and BloodMNIST has eight classes. Table I summarizes the splits.

A data-quality audit found no non-finite pixels, invalid labels, or constant images. The audit also detected 45 exact pixel-hash duplicate groups across the loaded benchmark arrays. We keep the official MedMNIST splits rather than modifying the benchmark after inspection and discuss this as a limitation.

TABLE I  
DATASETS USED IN THE FINAL STUDY

| Dataset        | Train | Val. | Test | Classes |
|----------------|-------|------|------|---------|
| PneumoniaMNIST | 4708  | 524  | 624  | 2       |
| BreastMNIST    | 546   | 78   | 156  | 2       |
| BloodMNIST     | 11959 | 1712 | 3421 | 8       |

### B. Frozen Image Representation

All main experiments use an ImageNet-pretrained ResNet-18 with its final classification layer removed [18]. Images are resized to  $96 \times 96$ . Grayscale images are copied to three channels and normalized with ImageNet statistics. The backbone is frozen and produces a 512-dimensional feature vector for each image. The feature vectors are cached before federated training.

This makes the experiment computationally lightweight, but it is important to state the scope correctly: the study simulates federated optimization over cached image features. It is not an end-to-end decentralized medical imaging system.

### C. Trainable Model

The trainable classifier contains a shared adapter followed by a linear head. The adapter is

$$512 \rightarrow 128 \rightarrow \text{LayerNorm} \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.15). \quad (1)$$

The output layer maps the 128-dimensional representation to the task classes. On the binary task, the adapter has 65,920 parameters and the classifier head has only 258 parameters, so the personalized head is about 0.39% of the trainable adapter-plus-head model.

### D. Non-IID Client Simulation

We simulate  $K = 5$  clients. For each class  $c$ , a client allocation profile is sampled as

$$\pi_c \sim \text{Dirichlet}(\alpha \mathbf{1}_K). \quad (2)$$

The main setting uses  $\alpha = 0.3$ . Smaller  $\alpha$  creates stronger class imbalance across clients. The same class-to-client profile is used to allocate the official train, validation, and test splits independently.

The training split has no forced class coverage. To make per-client multiclass evaluation possible, validation and test allocation gives each client one mandatory example from every class before distributing the remaining samples.

Partitions are additionally drawn by rejection sampling. A candidate profile is discarded and resampled if any client receives fewer than a dataset-specific minimum number of training samples (40 for PneumoniaMNIST, 25 for BreastMNIST, and 80 for BloodMNIST), or fewer than eight validation or test samples. On the development seed, the accepted profile was the third attempt for PneumoniaMNIST, the tenth for BreastMNIST, and the first for BloodMNIST. This constraint keeps every client trainable and every per-client metric computable, but it removes the most extreme allocations from

TABLE II  
SELECTED HYPERPARAMETERS

| Dataset        | FedProx $\mu$ | FT lr | FT ep. | Ditto $\lambda$ | PH $\mu$ | PH lr |
|----------------|---------------|-------|--------|-----------------|----------|-------|
| PneumoniaMNIST | 0.01          | 1e-3  | 5      | 1.0             | 0.1      | 2e-3  |
| BreastMNIST    | 0.001         | 1e-3  | 3      | 1.0             | 0.01     | 5e-4  |
| BloodMNIST     | 0.01          | 1e-3  | 5      | 0.1             | 1.0      | 5e-4  |

the sampled distribution. We therefore do not claim that the evaluation partitions are completely unconstrained Dirichlet draws.

### E. Optimization and Validation

The final configuration uses 20 global rounds, one local epoch per round, batch size 64, AdamW, adapter learning rate  $10^{-3}$ , weight decay  $10^{-4}$ , and class-weighted cross-entropy. WarmFedPer and ProxHead add 10 personalization rounds after the selected FedAvg checkpoint.

Hyperparameters are selected independently for each dataset using development seed 42 and validation data only. The search grid covers FedProx  $\mu \in \{0.001, 0.01, 0.1\}$ , ProxHead  $\mu \in \{0.01, 0.1, 1.0\}$ , head learning rates  $\{5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$ , Head-FT epochs  $\{1, 3, 5\}$ , and Ditto  $\lambda \in \{0.01, 0.1, 1.0\}$ . The selection rule is lexicographic: (1) bottom-two Macro-F1, (2) worst-client Macro-F1, and (3) mean Macro-F1. The selected values are given in Table II. On BloodMNIST the selected ProxHead  $\mu = 1.0$  lies at the upper boundary of the search grid, so a stronger penalty may be preferable on that dataset than anything we tested. Fresh confirmation seeds are not used for tuning. We use 10 confirmation seeds for PneumoniaMNIST and 5 each for BreastMNIST and BloodMNIST.

For ProxHead, the code uses a tensor-normalized head penalty:

$$\mathcal{L}_i = \mathcal{L}_{\text{CE},i} + \frac{\mu}{2} \sum_{\ell \in \mathcal{H}} \frac{1}{d_\ell} \|\theta_{i,\ell} - \theta_{g,\ell}\|_2^2, \quad (3)$$

where  $\mathcal{H}$  is the set of head parameter tensors and  $d_\ell$  is the number of elements in tensor  $\ell$ . The same normalized squared-distance implementation is used for the FedProx-style and Ditto-style baselines. For this reason we describe them as style-matched implementations rather than exact reproductions of the original penalty scaling.

### F. Evaluation Metrics

For a client metric  $M_k$ , we report the mean and the weakest-client value. We also define the bottom-two score as

$$M_{B2} = \frac{1}{2} \sum_{k \in B_2} M_k, \quad (4)$$

where  $B_2$  contains the two clients with the lowest score.

## IV. RESULTS

### A. Main Confirmation Results

Table III gives the final Macro-F1 comparison. Fig. 1 is the corresponding figure produced by the executed notebook.

No method wins every dataset. This is an important result rather than a problem with the experiment. On PneumoniaMNIST, Ditto-style training has the highest mean Macro-F1, while WarmFedPer has the best worst-client and bottom-two scores. On BreastMNIST, FedPer has the highest mean Macro-F1 and Head-FT is strongest on the lower tail. On BloodMNIST, Head-FT has the best mean and worst-client Macro-F1, while ProxHead has the highest bottom-two score by a small margin.

Two sets of rows in Table III are numerically indistinguishable on BreastMNIST, and we explain both rather than leave them to be read as errors. FedProx-style matches FedAvg because the selected  $\mu = 0.001$  makes the proximal gradient negligible against the cross-entropy gradient on this small dataset; the two runs diverge only in the fourth decimal of worst-client expected calibration error, confirming that the penalty is applied but has no material effect. ProxHead matches WarmFedPer on worst-client and bottom-two Macro-F1 for the same reason at  $\mu = 0.01$ : the paired difference is exactly zero on all five seeds. Both observations are consistent with the wider finding that proximal constraints matter only when they are strong enough relative to the data-fitting term.

#### B. Why WarmFedPer Is the Overall Project Selection

Although WarmFedPer does not win every individual column, it is the most consistent method by the project’s primary lower-tail criterion. Its bottom-two Macro-F1 exceeds FedAvg by 0.1004 on PneumoniaMNIST, 0.0811 on BloodMNIST, and 0.0205 on BreastMNIST.

These three numbers are not equally strong, and we state the difference explicitly. The PneumoniaMNIST and BloodMNIST gains are large relative to seed dispersion and are consistent in direction across every paired seed. The BreastMNIST gain is not: the paired analysis shows a median difference of exactly zero, with only one of five seeds separating the two methods and a bootstrap interval whose lower bound is zero. We therefore treat BreastMNIST as showing no detectable difference between WarmFedPer and FedAvg rather than as a third confirmation.

We also rank the seven methods separately on each dataset using bottom-two Macro-F1 and then average the three ranks. WarmFedPer obtains the best average rank (2.50), followed by Head-FT (2.67) and ProxHead (2.83). These three leading methods are separated by less than half a rank position over only three datasets, so this ordering indicates comparable overall performance rather than a demonstrated advantage. We select WarmFedPer as the most consistent of the three, not as the demonstrably best.

This result also explains the role of the WarmFedPer ablation. Warm-starting from FedAvg is itself a strong intervention. ProxHead is lower than WarmFedPer on PneumoniaMNIST, tied on BreastMNIST bottom-two Macro-F1, and only slightly higher on BloodMNIST. The proximal penalty therefore cannot be claimed as the main source of the overall improvement. This comparison is conservative with respect to ProxHead: WarmFedPer inherits the head learning rate that was selected

for ProxHead on validation data, so the ablation runs at a hyperparameter tuned for its comparator. That WarmFedPer nonetheless matches or exceeds ProxHead on the lower tail across all three datasets makes the conclusion harder to attribute to tuning advantage.

#### C. Client Dispersion

Lower-tail level and client dispersion are distinct quantities, and a method can raise the former without compressing the latter. Table IV reports the standard deviation of client Macro-F1 and the best-to-worst gap on PneumoniaMNIST, the dataset with ten confirmation seeds.

ProxHead gives the smallest dispersion on both measures, reducing the best-to-worst gap from 0.3748 under FedAvg to 0.2668 and the client standard deviation from 0.1370 to 0.0966. WarmFedPer is second on both. This is a different ordering from the lower-tail level results, where WarmFedPer leads and ProxHead does not, and it clarifies the role of the head penalty: the proximal constraint does not lift the weakest clients highest, but it does produce the most uniform client outcomes.

The effect is not universal. On BloodMNIST, the methods with the strongest lower tail also have the widest spread: Head-FT raises bottom-two Macro-F1 to 0.5833 while increasing the client gap from 0.1843 to 0.2176. Raising the floor and compressing the distribution are therefore separable objectives in this simulation, and reporting only one of them would misrepresent what personalization achieves.

#### D. Effect of Stronger Heterogeneity

To test whether the behavior changes with client skew, we run a separate PneumoniaMNIST experiment at  $\alpha \in \{0.1, 0.3, 1.0\}$  using three seeds per setting. Table V and Fig. 2 summarize the results.

At the strongest tested heterogeneity ( $\alpha = 0.1$ ), ProxHead gives the highest mean and worst-client Macro-F1, while FedPer is slightly higher on bottom-two Macro-F1. At  $\alpha = 0.3$ , WarmFedPer and FedPer are strongest. At  $\alpha = 1.0$ , FedPer is best across all three Macro-F1 summaries. This supports a cautious interpretation: the proximal head constraint may be more useful when client distributions are very skewed, but it can also restrict useful personalization when heterogeneity is milder. Each cell in Table V averages only three seeds and we do not report dispersion, so the within- $\alpha$  orderings are indicative rather than established; the differences between the three personalized methods at any single  $\alpha$  are small relative to the seed variation observed in Table III.

#### E. Representation Sanity Check

The main study uses ImageNet-pretrained frozen ResNet-18 features. We test this choice on PneumoniaMNIST with three independent seeds by comparing ImageNet features, a frozen randomly initialized ResNet-18, and raw normalized pixels. ImageNet features achieve mean/worst/bottom-two Macro-F1 of 0.4246/0.1113/0.2236. Random features achieve 0.3289/0.0382/0.1512, and raw pixels achieve

TABLE III  
FRESH CONFIRMATION MACRO-F1 (MEAN  $\pm$  STANDARD DEVIATION ACROSS SEEDS)

| Method        | PneumoniaMNIST                      |                                     |                                     | BreastMNIST                         |                                     |                                     | BloodMNIST                          |                                     |                                     |
|---------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
|               | Mean                                | Worst                               | Bottom-2                            | Mean                                | Worst                               | Bottom-2                            | Mean                                | Worst                               | Bottom-2                            |
| FedAvg        | .6304 $\pm$ .0924                   | .4309 $\pm$ .1261                   | .4870 $\pm$ .1221                   | .4854 $\pm$ .1153                   | .3023 $\pm$ .1405                   | .4023 $\pm$ .1022                   | .5721 $\pm$ .0232                   | .4748 $\pm$ .0566                   | .5034 $\pm$ .0332                   |
| FedProx-style | .6288 $\pm$ .1056                   | .4585 $\pm$ .1585                   | .4964 $\pm$ .1563                   | .4854 $\pm$ .1153                   | .3023 $\pm$ .1405                   | .4023 $\pm$ .1022                   | .5651 $\pm$ .0296                   | .4730 $\pm$ .0596                   | .5047 $\pm$ .0380                   |
| FedPer        | .6977 $\pm$ .0813                   | .5359 $\pm$ .0995                   | .5781 $\pm$ .0989                   | <b>.5351 <math>\pm</math> .0743</b> | .4423 $\pm$ .0335                   | .4689 $\pm$ .0373                   | .5366 $\pm$ .0604                   | .4452 $\pm$ .0881                   | .4718 $\pm$ .0758                   |
| Head-FT       | .6940 $\pm$ .0691                   | .5071 $\pm$ .0862                   | .5550 $\pm$ .0906                   | .5263 $\pm$ .1102                   | <b>.4518 <math>\pm</math> .1061</b> | <b>.4776 <math>\pm</math> .0837</b> | <b>.6703 <math>\pm</math> .0625</b> | <b>.5659 <math>\pm</math> .0636</b> | .5833 $\pm$ .0616                   |
| Ditto-style   | <b>.7016 <math>\pm</math> .0657</b> | .5047 $\pm$ .0469                   | .5523 $\pm$ .0745                   | .5257 $\pm$ .0665                   | .4081 $\pm$ .0930                   | .4438 $\pm$ .0490                   | .5041 $\pm$ .0204                   | .3827 $\pm$ .0423                   | .4260 $\pm$ .0284                   |
| WarmFedPer    | .6981 $\pm$ .0742                   | <b>.5385 <math>\pm</math> .1088</b> | <b>.5874 <math>\pm</math> .0971</b> | .4924 $\pm$ .1310                   | .3225 $\pm$ .1857                   | .4228 $\pm$ .1480                   | .6570 $\pm$ .0470                   | .5530 $\pm$ .0734                   | .5845 $\pm$ .0615                   |
| ProxHead      | .6686 $\pm$ .0960                   | .5228 $\pm$ .1087                   | .5676 $\pm$ .1023                   | .4939 $\pm$ .1344                   | .3225 $\pm$ .1857                   | .4228 $\pm$ .1480                   | .6605 $\pm$ .0481                   | .5326 $\pm$ .0446                   | <b>.5862 <math>\pm</math> .0365</b> |

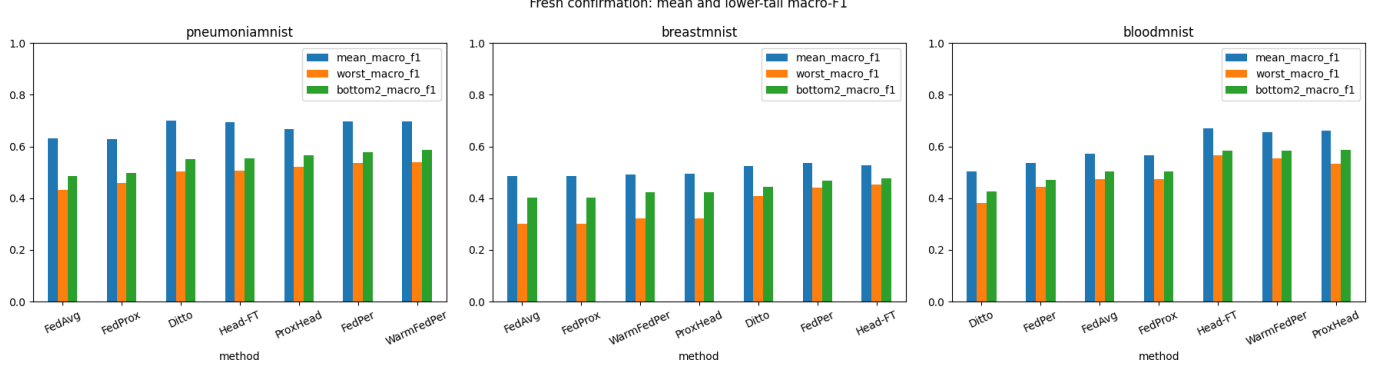


Fig. 1. Each panel shows mean, worst-client, and bottom-two Macro-F1 for the seven methods.

TABLE IV  
PNEUMONIAMNIST CLIENT DISPERSION (LOWER IS MORE UNIFORM)

| Method        | Client SD of Macro-F1 | Best–Worst Gap |
|---------------|-----------------------|----------------|
| FedAvg        | .1370                 | .3748          |
| FedProx-style | .1260                 | .3265          |
| FedPer        | .1164                 | .3184          |
| Head-FT       | .1343                 | .3553          |
| Ditto-style   | .1386                 | .3541          |
| WarmFedPer    | .1086                 | .2988          |
| ProxHead      | <b>.0966</b>          | <b>.2668</b>   |

TABLE V  
PNEUMONIAMNIST HETEROGENEITY SENSITIVITY

| $\alpha$ | Method     | Mean F1      | Worst F1     | Bottom-2 F1  |
|----------|------------|--------------|--------------|--------------|
| 0.1      | FedAvg     | .4973        | .2865        | .2964        |
|          | FedPer     | .5815        | .4106        | <b>.4849</b> |
|          | WarmFedPer | .5973        | .4397        | .4712        |
|          | ProxHead   | <b>.6044</b> | <b>.4426</b> | .4799        |
| 0.3      | FedAvg     | .6643        | .4508        | .5214        |
|          | FedPer     | .7192        | <b>.5974</b> | .6256        |
|          | WarmFedPer | <b>.7246</b> | .5962        | <b>.6265</b> |
|          | ProxHead   | .7180        | .5940        | .6106        |
| 1.0      | FedAvg     | .7629        | .6148        | .6696        |
|          | FedPer     | <b>.8120</b> | <b>.7347</b> | <b>.7556</b> |
|          | WarmFedPer | .7969        | .7013        | .7264        |
|          | ProxHead   | .7945        | .7063        | .7255        |

0.3309/0.0382/0.1562. These runs use three partition seeds (302, 312, 322) that are disjoint from the confirmation seeds, so the absolute values are not comparable with Table III; the ImageNet standard deviation across these three seeds is

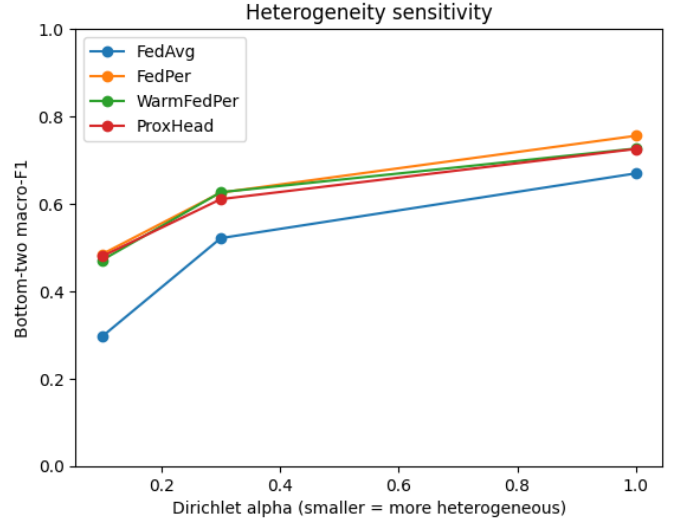


Fig. 2. Bottom-two Macro-F1 on PneumoniaMNIST at different Dirichlet concentrations. Smaller  $\alpha$  creates stronger heterogeneity.

0.2118, which illustrates how strongly a single Dirichlet draw determines task difficulty. Only the within-experiment ordering is meaningful, and that ordering places ImageNet features ahead of both alternatives on all three summaries. Random convolutional features and raw pixels are also nearly identical to each other, which indicates that the benefit comes from pretraining rather than from the convolutional architecture alone.

### F. Calibration and Client Diagnostics

Personalization also changes how well client confidence matches client accuracy. On PneumoniaMNIST, worst-client expected calibration error is 0.3916 under FedAvg. Every personalized method reduces it: 0.2714 for ProxHead, 0.2248 for WarmFedPer, 0.1973 for FedPer, and 0.1486 for Ditto-style training. Mean-client ECE follows the same ordering, falling from 0.1903 under FedAvg to 0.0781 under Ditto. The ordering here is close to the inverse of the lower-tail Macro-F1 ordering, where Ditto is weakest: methods that keep more parameters local produce better-calibrated clients even when they do not produce more accurate ones. For a deployment in which a weak client’s confidence would inform downstream decisions, this is a second respect in which the global model serves that client poorly.

The notebook also measures NLL, Brier score, and client-support variables. On the 50 PneumoniaMNIST confirmation clients, none of the tested Spearman associations between personalization gain and training size, minority support, majority ratio, or label entropy reaches  $p < 0.05$ . The largest trend for ProxHead is Jensen–Shannon divergence from the global label distribution ( $\rho = 0.2764$ ,  $p = 0.0520$ ). We therefore do not claim a proven mechanism for why some clients benefit more than others.

A prespecified weak-client case study showed that personalized methods can change minority- and majority-class errors in different ways even when overall accuracy is similar.

### V. DISCUSSION

Three points summarize the results.

**First, personalization should not be judged only by the average.** The development observation that motivated this project was that FedPer could improve the mean while hurting a weak client. The final study therefore reports lower-tail metrics directly. This is especially important under strong label imbalance, where high accuracy can coexist with weaker Macro-F1. BloodMNIST provides a clear illustration: Ditto-style training reaches 0.8922 mean accuracy, the second highest of the seven methods, but its mean Macro-F1 of 0.5041 is the lowest. A single accuracy figure would have concealed how poorly that model serves the minority classes on which weaker clients depend.

**Second, warm initialization is a major part of the benefit.** WarmFedPer starts from a trained FedAvg solution and then allows only a small client-specific head to personalize while the adapter continues to be shared. It obtains the best average lower-tail rank and improves the FedAvg bottom-two score on PneumoniaMNIST and BloodMNIST, with no detectable difference on BreastMNIST. This is a simple result and does not require claiming a new general FL algorithm.

**Third, stronger regularization is conditional.** ProxHead improves over FedAvg on PneumoniaMNIST and BloodMNIST and is particularly competitive under  $\alpha = 0.1$ . However, it does not consistently beat its compute-matched WarmFedPer ablation. The experiment therefore does not support the claim that a proximal head penalty is always beneficial. A more

defensible conclusion is that stronger anchoring can help under severe heterogeneity but may be unnecessary under milder client skew.

This also clarifies the project’s novelty. FedAvg, FedProx, FedPer, and Ditto are established methods. WarmFedPer is a project ablation that isolates warm-start personalization. ProxHead is a lightweight project extension that restricts proximal control to the classifier head. The main contribution is the controlled lower-tail analysis and the resulting empirical finding, not the invention of proximal personalization itself.

### VI. LIMITATIONS

The results should be interpreted within the following limits.

- The five clients are simulated data partitions, not real hospitals. Dirichlet label skew does not represent scanner, demographic, geographic, workflow, or annotation shifts.
- Client partitions are drawn by rejection sampling under minimum per-client size constraints, and validation and test partitions force one example of each class per client before the remaining allocation. This keeps every client trainable and every per-client metric computable, but it removes the most extreme allocations and so departs from a pure Dirichlet split.
- The frozen ResNet-18 features are computed centrally in the simulation. The project does not implement end-to-end decentralized image training.
- No secure aggregation, differential privacy, encryption, or other formal privacy mechanism is implemented.
- Lower-tail client performance is not demographic fairness. The simulated clients have no protected-group meaning.
- MedMNIST is a lightweight research and educational benchmark and is explicitly not intended for clinical use [17].
- The audit found 45 exact pixel-hash duplicate groups in the loaded benchmark arrays, and some of these groups span more than one official split. We retain the official MedMNIST split, so this remains a benchmark-level limitation for interpreting absolute scores.
- BreastMNIST is small, and BreastMNIST and BloodMNIST use only five confirmation seeds. This limits statistical power.
- The implemented proximal penalty uses a tensor-normalized mean-squared distance, so the FedProx-style and Ditto-style implementations are not exact reproductions of the original penalty scale.

### VII. CONCLUSION

This project asks whether lightweight personalization can improve weaker clients under simulated non-IID medical data. Across three MedMNIST tasks, WarmFedPer gives the best average cross-dataset rank by bottom-two Macro-F1 and improves the FedAvg lower tail on two of the three datasets, with the BreastMNIST difference falling within seed noise. However, no method is best everywhere. Head-FT is strongest

on several BreastMNIST and BloodMNIST metrics, and Prox-Head is competitive under severe heterogeneity and gives the best BloodMNIST bottom-two Macro-F1.

The final takeaway is therefore not that one custom method universally wins. The evidence supports a simpler conclusion: starting personalization from a good global model and keeping the personalized component small is a strong overall strategy in this simulation. Additional proximal regularization can help in harder non-IID conditions, but its benefit depends on the client distribution.

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